Q1

1. yeduc 's coefficient is 17 826.42 std. err. is 193.9603
nin 's coefficient is -6880.068 std. err is. 714.78

For comment on statistical significance: both predictors are statistically significant at 1% level since their p value < 0.01

R^2: $R^2 = 0.1238$, it shows that education and number of school explain 12.38% of variances in wages, there are other variables such as work experiences may play a significant role.

2. The misspecified model omit the variable nin, the model could produce biased estimate for β_1

Using STATA, the correlation of yeduc and nin is -0.0624 which is smaller than 0, since $\beta_2 < 0$, so the nin variable will produce positive bias, if omit the nin, it will have negative biase to education.

3. Based on estimates and comparison. the direction of cov(yeduc, nin) may be negative correlation.

Based on dataset, correlation of (yeduc, nin) = -0.0621 < 0

so two results consistent with each other

4. model3:

$wage_{ij} = \alpha + \beta_1 education_i + \beta_2 nin_j + \beta_3 en71 + \mu_i$

5.

$H_0: \beta_3 = 0$ suppose $\alpha = 0.05$

$H_1: \beta_3 \neq 0$

from STATA, we get $t = 3.76 > 1.96$

so we reject H_0,

the enrollment rate in 1971 also change the wage outcome.

6. We shall use F test.

$H_0: \beta_2 = 0$ and $\beta_3 = 0$ suppose $\alpha = 0.05$

$H_1: \beta_2 \neq 0$ or $\beta_3 \neq 0$

By STATA, we get $\chi^2 = 95.16$

$Prob > \chi^2 = 0.0000$, so under 0.05 significant level, we reject H_0.

So the effects of "nin" and "en71" are significant

Q2

1.

	yedu	nin	constant
Linear probability model	0.0359757	-0.133587	-0.001791
	(0.0002298)	(0.000836)	(0.0027973)
Probit model	0.1431199	-0.0528456	-2.020965
	(0.0010296)	(0.0036573)	(0.0131103)

for each additional unit increase in yedu, the probability of being a high-wage individual increases by about 3.60%

2.

1. Predicted Probabilities outside [0,1]

Range: The LPM can predict probabilities outside [0,1], which is not meaningful for binary outcome

2. Non-Normal Error Distribution:

The errors in LPM not follow a normal distribution, leading to insufficient estimates and potentially biased standard errors

3.

For $X=8$

$$P(Y|X=8, Z=3) = \Phi(\beta_0 + \beta_1 8 + \beta_2 3)$$

For $X=12$

$$P(Y|X=12, Z=3) = \Phi(\beta_0 + \beta_1 12 + \beta_2 3)$$

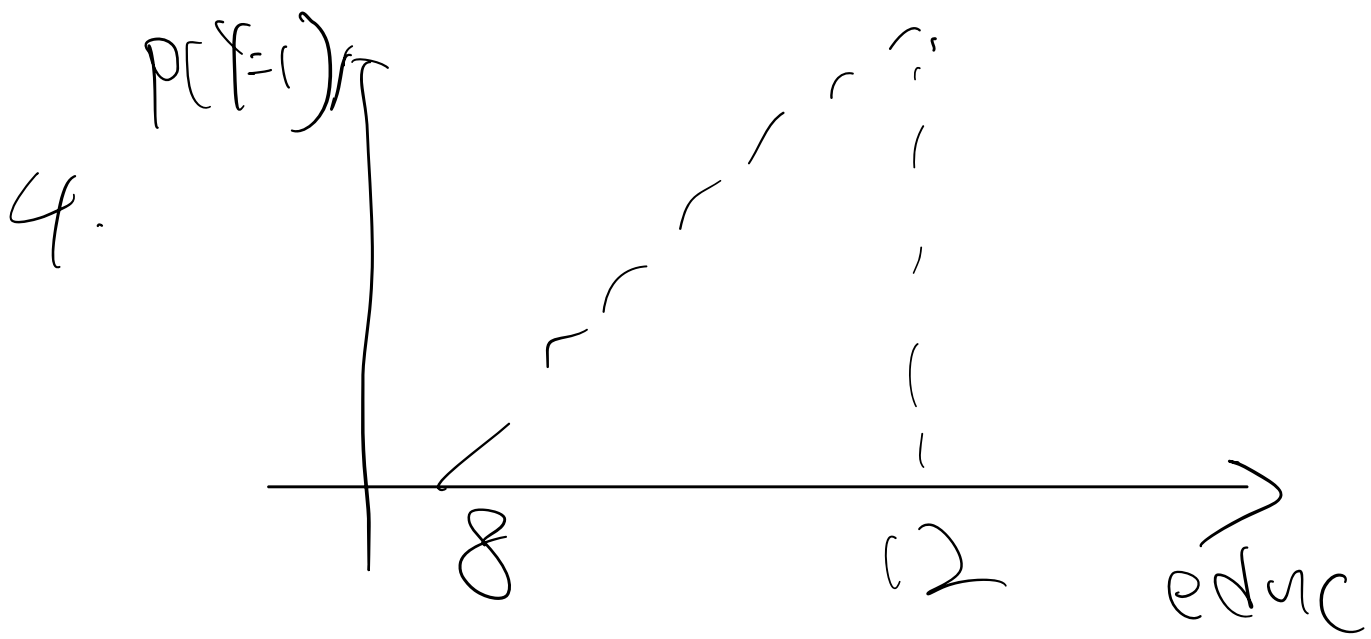
The change in probability is:

$$\Delta P = P(Y|X=12, Z=3) - P(Y|X=8, Z=3)$$

$$= \Phi(\beta_0 + 0.143120 \cdot 12 - 0.052840 \cdot 3) -$$

$$\Phi(\beta_0 + 0.143120 \cdot 8 - 0.052840 \cdot 3)$$

4.



Q3

1- coefficient of high level is 112764.3
This means, individuals in high-education group (high_level=1) earn on average $112764.3 more than low-education group (high-level=0), holding all else constant.

2. 1) $wage = \beta_0 + \beta_1 high_level + \beta_2 urban + \beta_3 (high_level \times urban) + \varepsilon$

$\beta_0 = 152489$ (1446.661) → 1446.661 is std-err

$\beta_1 = 99802.76$ (1914.018)

$\beta_2 = 25014.22$ (3553.394)

$\beta_3 = 2341.8$ (4131.893)

2) H0: $\beta_3 = 0$

H1: $\beta_3 \neq 0$

The p-value for interaction term is 0.0000. so p-value less than any typical significance level (α=0.05, e.g)

so we reject null hypothesis

so Interaction between high education level and urban birthplace is statistically significant.